INTERACTIVE SIMULATION TEACHING HANDBOOK

Supply Chain Finance & Operations Analytics

Instructor Guide with Equations, Worked Examples, and Exercises

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Overview

This handbook accompanies an integrated browser-based simulation dashboard designed for graduate and advanced undergraduate courses in supply chain management, financial operations, enterprise risk management, and applied business analytics. The dashboard consolidates three interactive modules into a single HTML file that runs entirely in the browser with no server infrastructure or software installation required.

Each module is grounded in a formal analytical model. This handbook provides the complete mathematical derivations, defines all notation, walks through worked numerical examples with reference to specific dashboard configurations, and offers structured classroom exercises with discussion prompts.

Module	Topic	Session
1	Cash Conversion Cycle & Working Capital	75–90 min
2	Dynamic Discounting & Reverse Factoring	75–90 min
3	Monte Carlo Profit Risk Simulation	90–120 min

Contents

Overview **2**

1 Cash Conversion Cycle and Working Capital **3**
- 1.1 Dashboard Overview . . . 3
- 1.2 Learning Objectives . . . 3
- 1.3 Notation . . . 4
- 1.4 Core Equations . . . 4
- 1.5 Reading the Timeline Chart . . . 4
- 1.6 Reading the Sensitivity Chart . . . 5
- 1.7 Worked Example . . . 5
- 1.8 Classroom Exercises . . . 5

2 Dynamic Discounting and Reverse Factoring **7**
- 2.1 Dashboard Overview . . . 7
- 2.2 Learning Objectives . . . 7
- 2.3 Notation . . . 7
- 2.4 Core Equations . . . 8
  - 2.4.1 Dynamic Discounting . . . 8

2.4.2 Win-Win Condition (Credit Arbitrage) . . . 8
2.4.3 Reverse Factoring . . . 8
2.5 Reading the Discount Curves Chart . . . 8
2.6 Reading the Net Benefit Chart . . . 9
2.7 Worked Example . . . 9
2.8 Classroom Exercises . . . 9

3 Monte Carlo Profit Risk Simulation **11**
3.1 Dashboard Overview . . . 11
3.2 Learning Objectives . . . 11
3.3 Notation . . . 12
3.4 Core Equations . . . 12
3.4.1 Profit Function . . . 12
3.4.2 Optimal Order Quantity . . . 12
3.4.3 Risk Metrics from Simulation . . . 12
3.5 Reading the Profit Histogram . . . 13
3.6 Reading the $\mathbb{E}[\pi]$ and VaR Curves . . . 13
3.7 Worked Example . . . 13
3.8 Classroom Exercises . . . 14

4 Curriculum Alignment **15**
4.1 Mapping to BU Administrative Sciences . . . 15
4.2 AI and Machine Learning Extensions . . . 15
4.3 Technical Notes . . . 15
4.4 Suggested Readings . . . 15

1 Cash Conversion Cycle and Working Capital

1.1 Dashboard Overview

The Module 1 interface consists of a left panel with six parameter sliders (Annual Revenue, COGS%, DIO, DSO, DPO, and Cost of Capital) and a right panel rendering two real-time visualizations: the operating cycle timeline and the CCC-versus-financing-cost sensitivity chart. The top stats bar displays the computed CCC, operating cycle length, net working capital, annual financing cost, and the marginal value of a one-day CCC reduction.

Default configuration: Revenue = \$80M, COGS = 60%, DIO = 45d, DSO = 50d, DPO = 25d, CoC = 6%. CCC = 70 days, NWC = \$13.59M, Financing Cost = \$815K/yr.

1.2 Learning Objectives

Learning Objectives

After completing this module, students should be able to:

1. Compute the CCC from financial statement data and interpret each component (DIO, DSO, DPO) in operational terms
2. Calculate the net working capital requirement and annual financing cost implied by a given CCC
3. Evaluate the financial impact of targeted improvements in any single CCC component
4. Explain why firms such as Amazon and Dell have achieved negative CCC values

1.3 Notation

D	Annual revenue, measured in dollars
$COGS$	Cost of goods sold, annual, measured in dollars
DIO	Days Inventory Outstanding: average number of days inventory is held before sale
DSO	Days Sales Outstanding: average number of days to collect accounts receivable
DPO	Days Payable Outstanding: average number of days the firm defers payment to suppliers
r	Weighted average cost of capital (WACC), expressed as an annual rate
CCC	Cash Conversion Cycle, measured in days
OC	Operating Cycle $= DIO + DSO$, measured in days
NWC	Net Working Capital required, measured in dollars

1.4 Core Equations

The cash conversion cycle is defined as the difference between the operating cycle and the payment deferral period:

$$CCC = DIO + DSO - DPO \tag{1}$$

Each component is derived from the balance sheet and income statement:

$$DIO = \frac{\text{Average Inventory}}{COGS} \times 365, \quad DSO = \frac{\text{Average AR}}{D} \times 365, \quad DPO = \frac{\text{Average AP}}{COGS} \times 365 \tag{2}$$

Net working capital:

$$NWC = \frac{COGS}{365} \times DIO + \frac{D}{365} \times DSO - \frac{COGS}{365} \times DPO \tag{3}$$

The annual financing cost and marginal value of a one-day CCC reduction:

$$\text{Annual Financing Cost} = NWC \times r, \qquad \Delta\text{Cost per Day} = \frac{D}{365} \times r \tag{4}$$

1.5 Reading the Timeline Chart

The timeline chart visualizes the operating cycle as three horizontal bars on a shared time axis measured in days from the date of inventory purchase. The blue bar represents DIO, the orange

bar represents DSO (starting at day DIO and extending to day $DIO + DSO$), and the green bar represents DPO. The red dashed bracket beneath identifies the CCC as the gap between the end of DPO and the end of the operating cycle. When DPO exceeds $DIO + DSO$, the CCC turns negative and the annotation changes to indicate float generation.

1.6 Reading the Sensitivity Chart

The sensitivity chart plots annual financing cost on the vertical axis against CCC in days on the horizontal axis. Because financing cost equals daily revenue times the cost of capital times CCC, the relationship is linear. The current operating point is marked with a red dot. The green-shaded region to the left of $CCC = 0$ represents configurations where the firm earns float income rather than paying financing costs. This chart helps students see that each day of CCC reduction saves a fixed dollar amount, regardless of whether the starting CCC is 100 days or 10 days.

1.7 Worked Example

Worked Example: Large Manufacturer (Revenue = \$80M, COGS = 60%)

Step 1: Set Revenue = \$80M, COGS = 60% of revenue = \$48M. Set $DIO = 45$, $DSO = 50$, $DPO = 25$.
Step 2: $CCC = 45 + 50 - 25 = 70$ days. Operating Cycle $= 45 + 50 = 95$ days.
Step 3: Inventory = (\$32.5M/365) × 35 = \$3,116,438. AR = (\$50M/365) × 40 = \$5,479,452. AP = (\$32.5M/365) × 30 = \$2,671,233.
Step 4: NWC = \$3,116,438 + \$5,479,452 − \$2,671,233 = \$5,924,657.
Step 5: At $r = 6\%$: Annual Financing Cost = \$13,589,041 × 0.06 = \$815,342.
Step 6: One-day CCC reduction value: (\$80M/365) × 0.06 = \$13,151/year.
Step 7: Try adjusting DPO from 25 to 70 in the dashboard. CCC drops from 70 to 25 days, NWC falls by approximately \$5.92M, and financing cost drops by about \$355K. This demonstrates how extending payment terms frees substantial working capital.

1.8 Classroom Exercises

Exercise 1: DSO vs. DPO Levers

A 10-day DSO reduction and a 10-day DPO increase both reduce CCC by 10 days. However, the NWC impact differs because DSO is computed on revenue while DPO is computed on COGS. Using the default parameters (COGS = 65% of revenue), compute the exact dollar difference.
Solution: The DSO reduction frees \$50M/365 × 10 = \$1,369,863, while the DPO extension frees \$32.5M/365 × 10 = \$890,411. The DSO lever releases 54% more capital.

Exercise 2: Negative CCC

Set revenue to \$100M and DPO to 120 days while keeping $DIO = 35$ and $DSO = 40$. Observe that CCC becomes negative (-45 days). Discuss: Amazon's CCC was approximately -30 days in recent years. What business model features enable this?
Discussion points: Consignment inventory, credit card settlement in 1–2 days, 60–90 day supplier payment terms. Is this sustainable or exploitative?

Exercise 3: Interest Rate Sensitivity

Vary the cost of capital from 2% to 15%. Observe how the sensitivity line steepens. For a high-interest-rate environment (e.g., emerging market at 12%), the same CCC of 45 days costs \$739K/year instead of \$296K. Discuss: How should a CFO in a high-rate environment prioritize CCC reduction differently from one in a low-rate environment?

2 Dynamic Discounting and Reverse Factoring

2.1 Dashboard Overview

The Module 2 interface contains seven parameter sliders covering the invoice amount, payment terms, early payment day, discount APR, buyer's and supplier's costs of capital, and the bank factoring rate. Below the sliders, three cards compare the Standard, Dynamic Discounting, and Reverse Factoring strategies. The right panel shows the discount curves chart and the net benefit sensitivity chart. The top stats bar indicates whether a credit arbitrage exists.

Default configuration: Invoice = \$750K, Terms = 90d, Early = 15d, $r_d = 7\%$, $r_b = 4\%$, $r_s = 14\%$, $r_k = 3.5\%$. Win-win condition holds: $4\% < 7\% < 14\%$, combined surplus = \$20.8K.

2.2 Learning Objectives

Learning Objectives

After completing this module, students should be able to:

1. Compute the dollar discount and annualized return for a dynamic discounting arrangement
2. Evaluate when dynamic discounting generates a positive net return for the buyer versus when it does not
3. Compare dynamic discounting with bank-intermediated reverse factoring
4. Articulate the win-win condition as a credit arbitrage inequality

2.3 Notation

F	Invoice face value (dollars)
T	Standard payment terms (days from invoice date)
t	Early payment day (days from invoice date), where $t < T$
Δt	Acceleration window $= T - t$ (days of early payment)
r_d	Discount APR: annualized discount rate for early payment
r_b	Buyer's cost of capital (WACC), annualized
r_s	Supplier's cost of capital, annualized
r_k	Bank factoring rate in reverse factoring, annualized
d	Dollar discount on the invoice

2.4 Core Equations

2.4.1 Dynamic Discounting

The discount amount, buyer's opportunity cost, and net benefits:

$$d = F \times \frac{r_d}{365} \times (T - t) \tag{5}$$

$$\text{Buyer Net Benefit} = F \times \frac{r_d - r_b}{365} \times (T - t) \tag{6}$$

$$\text{Supplier Net Benefit} = F \times \frac{r_s - r_d}{365} \times (T - t) \tag{7}$$

2.4.2 Win-Win Condition (Credit Arbitrage)

Both parties benefit simultaneously if and only if:

$$\boxed{r_b \ < \ r_d \ < \ r_s} \tag{8}$$

The total surplus captured when this condition holds:

$$\text{Total Surplus} = F \times \frac{r_s - r_b}{365} \times (T - t) \tag{9}$$

2.4.3 Reverse Factoring

In reverse factoring, a bank pays the supplier early at a rate anchored to the buyer's credit rating:

$$\text{Bank Fee} = F \times \frac{r_k}{365} \times (T - t), \qquad \text{Supplier Net} = F \times \frac{r_s - r_k}{365} \times (T - t), \qquad \text{Buyer Net} = 0 \tag{10}$$

2.5 Reading the Discount Curves Chart

Four lines plot the dollar discount (or cost) as a function of early payment day, each at a different annualized rate: the supplier's CoC (r_s, red), the discount APR (r_d, dark blue), the buyer's CoC (r_b, light blue), and the bank rate (r_k, orange dashed). All lines converge to zero at the payment due date and fan out as the early payment day moves earlier. At any given day, the vertical distance between two lines represents the net benefit or cost differential.

2.6 Reading the Net Benefit Chart

The supplier and buyer net benefit curves are plotted as functions of the early payment day. When $r_b < r_d < r_s$, both curves are positive, visually confirming the win-win condition. The magnitude increases as the early payment day moves earlier (more days of acceleration).

2.7 Worked Example

Worked Example: \$750K Invoice, Net 90 (Win-Win Scenario)

Step 1: $F = \$500{,}000$, $T = 60$ days, $t = 10$ days, $\Delta t = 50$ days.
Step 2: $r_d = 7.0\%$, $r_b = 4.0\%$, $r_s = 14.0\%$, $r_k = 3.5\%$.
Step 3: Dynamic discount: $d = \$500{,}000 \times (0.03/365) \times 50 = \$2{,}055$.
Step 4: Buyer opportunity cost $= \$750{,}000 \times (0.04/365) \times 75 = \$6{,}164$. Buyer net $= \$10{,}788 - \$6{,}164 = +\$4{,}624$. The buyer earns a positive return because $r_d\,(7\%) > r_b\,(4\%)$.
Step 5: Supplier financing saved $= \$750{,}000 \times (0.14/365) \times 75 = \$21{,}575$. Supplier net $= \$21{,}575 - \$10{,}788 = \$10{,}788$. The supplier benefits because $r_d\,(7\%) < r_s\,(14\%)$.
Step 6: Win-win check: $r_b\,(4\%) < r_d\,(7\%) < r_s\,(14\%)$. The credit arbitrage condition holds. Total surplus $= \$4{,}624 + \$10{,}788 = \$15{,}411$.
Step 7: Reverse factoring: Bank fee $= \$750{,}000 \times (0.035/365) \times 75 = \$5{,}393$. Supplier net $= \$21{,}575 - \$5{,}393 = \$16{,}182$. Buyer net $= \$0$.
Step 8: Reverse factoring produces higher combined surplus (\$16,182 vs. \$15,411), but dynamic discounting is the only strategy where the buyer also captures value.
Step 9: Set $r_d = 3\%$ (below $r_b = 4\%$). The buyer net flips negative, demonstrating the arbitrage boundary in real time.

2.8 Classroom Exercises

Exercise 1: Finding the Win-Win Zone

Slowly increase the Discount APR slider from 1% to 15%. Observe the exact APR at which the buyer's net benefit transitions from negative to positive (it equals $r_b = 5\%$). Then find the APR at which the supplier's net benefit turns negative (it equals $r_s = 10\%$). The range 5%–10% is the win-win zone.

Exercise 2: Trade Credit vs. Reverse Factoring

A supplier is offered 2/10 net 30 (2% discount for payment on day 10 instead of day 30). Compute the annualized cost:

$$\frac{0.02}{0.98} \times \frac{365}{20} = 37.2\%$$

Compare this with reverse factoring at 4% APR. The trade credit discount is 9× more expensive, illustrating why SCF platforms are displacing traditional trade discounts.

Exercise 3: Cash-Rich Buyer Scenario

Set the buyer's CoC to 2% (cash-rich tech company) and supplier CoC to 15% (small manufacturer). Total surplus at day 10: $F \times (0.15 - 0.02)/365 \times 50 = \$8{,}904$. Discuss: Why might Apple, Google, and Microsoft run large-scale SCF programs?

Discussion: They have excess cash earning near-zero returns and suppliers who borrow at 10–15%.

3 Monte Carlo Profit Risk Simulation

3.1 Dashboard Overview

The Module 3 interface contains six parameter sliders (Mean Demand, Demand StdDev, Unit Cost, Selling Price, Salvage Value, and Order Quantity) plus a re-run button that triggers 10,000 fresh Monte Carlo trials. The right panel displays the profit distribution histogram (with VaR and expected profit annotated) and two smaller charts showing Expected Profit and VaR(5%) as functions of Q. The top stats bar reports $\mathbb{E}[\pi]$, $\sigma(\pi)$, VaR(5%), achieved service level, and the analytically optimal Q^*.

Default configuration: $\mu = 500$, $\sigma = 120$, $c = \$45$, $p = \$90$, $s = \$15$, $Q = 820$. *Results:* $\mathbb{E}[\pi] = \$12.9K$, *VaR(5%)*$= -\$1.8K$, $Q^* = 530$.

3.2 Learning Objectives

Learning Objectives

After completing this module, students should be able to:

1. Derive the Newsvendor critical ratio and compute Q^* for normally distributed demand
2. Interpret VaR(5%) as a downside risk threshold
3. Analyze the tradeoff between expected profit and tail risk as Q varies
4. Connect inventory risk to the mean-variance frontier concept from portfolio theory
5. Articulate how ML demand forecasting integrates with Monte Carlo simulation

3.3 Notation

p	Selling price per unit (dollars)
c	Unit procurement cost (dollars)
s	Salvage value per unsold unit (dollars), where $s < c$
D	Random demand, $D \sim \mathcal{N}(\mu, \sigma^2)$, truncated at zero
μ	Mean of the demand distribution
σ	Standard deviation of the demand distribution
Q	Order quantity (decision variable)
Q^*	Optimal order quantity maximizing $\mathbb{E}[\pi]$
CR	Critical ratio $= (p-c)/(p-s)$
$\pi(Q, D)$	Profit realization for given Q and demand D
$\Phi(\cdot)$	CDF of the standard normal distribution

3.4 Core Equations

3.4.1 Profit Function

$$\pi(Q, D) = p \cdot \min(Q, D) + s \cdot \max(Q - D,\, 0) - c \cdot Q \tag{11}$$

When $D \geq Q$: all units sell, $\pi = (p - c) \cdot Q$. When $D < Q$: $\pi = p \cdot D + s \cdot (Q - D) - c \cdot Q = (p - s) \cdot D - (c - s) \cdot Q$.

3.4.2 Optimal Order Quantity

The first-order condition yields the critical ratio, which is the target service level:

$$CR = \frac{p - c}{p - s} \tag{12}$$

$$\boxed{Q^* = \mu + \sigma \cdot \Phi^{-1}(CR)} \tag{13}$$

When $CR > 0.5$, the profit margin exceeds the overage cost, and $Q^* > \mu$. When $CR < 0.5$, $Q^* < \mu$.

3.4.3 Risk Metrics from Simulation

From $N = 10{,}000$ simulated profit realizations $\{\pi_1, \ldots, \pi_N\}$:

$$\mathbb{E}[\pi] = \frac{1}{N}\sum_{i=1}^{N}\pi_i, \qquad \sigma(\pi) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\pi_i - \mathbb{E}[\pi])^2} \tag{14}$$

$$\text{VaR}(5\%) = \text{5th percentile of } \{\pi_1, \ldots, \pi_N\}, \qquad \text{Sharpe-like Ratio} = \frac{\mathbb{E}[\pi]}{\sigma(\pi)} \tag{15}$$

3.5 Reading the Profit Histogram

The horizontal axis measures profit in dollars. Bins to the left of the VaR(5%) line (red dashed) are shaded red, representing the worst 5% of outcomes. The green dashed line marks $\mathbb{E}[\pi]$. With $Q = 820$ well above $Q^* = 530$, the distribution forms a near-perfect bell curve because Q is large enough that almost all demand is satisfied (SL = 99.5%), and profit variation comes mainly from unsold-unit salvage losses. The red-shaded tail shows profits below approximately −$1,800, meaning there is a 5% chance of actual financial loss.

3.6 Reading the $\mathbb{E}[\pi]$ and VaR Curves

The two smaller charts plot Expected Profit and VaR(5%) as functions of Q. The $\mathbb{E}[\pi]$ curve is concave, peaking at Q^*. The VaR curve initially increases (more stock means more sales in low-demand scenarios) but eventually decreases sharply (extreme overstocking produces large losses). Together, these charts form a visual risk-return frontier: **the Q that maximizes $\mathbb{E}[\pi]$ is not the same Q that maximizes VaR(5%)**.

3.7 Worked Example

Worked Example: Seasonal Product ($\mu = 500$, $p = \$90$, $c = \$45$, $s = \$15$)

Step 1: $\mu = 500$, $\sigma = 120$, $p = \$90$, $c = \$50$, $s = \$15$.
Step 2: $CR = (90 - 50)/(90 - 15) = 40/75 = 0.5333$.
Step 3: $\Phi^{-1}(0.600) \approx 0.253$. $Q^* = 500 + 120 \times 0.253 = 530$ units.
Step 4: At $Q = 820$ (well above Q^*): $\mathbb{E}[\pi] \approx \$12.9\text{K}$, VaR(5%)$\approx -\$1.8\text{K}$, SL$\approx$ 99.5%, Sharpe$\approx$ 1.44. The histogram is a symmetric bell curve.
Step 5: At $Q = Q^* = 530$: $\mathbb{E}[\pi] \approx \$16.8\text{K}$, VaR(5%)$\approx$ \$6.5K, Sharpe$\approx$ 2.9. Higher expected profit *and* positive VaR. Right-skewed histogram.
Step 6: At $Q = 300$ (conservative): $\mathbb{E}[\pi] \approx \$12.5\text{K}$, VaR(5%)$\approx$ \$11.8K, Sharpe$\approx$ 7.2. Nearly all scenarios yield stable profit, but upside is capped.

Three regimes emerge: $Q \ll Q^*$ (safe but low upside), $Q \approx Q^*$ (max expected profit, moderate risk), and $Q \gg Q^*$ (symmetric bell but negative VaR). The choice depends on the manager's risk tolerance.

3.8 Classroom Exercises

Exercise 1: Perishable Goods ($s = 0$)

Set salvage value to \$0 (perishable goods). Observe how the histogram shifts left and VaR(5%) drops substantially. Compute the new Q^*: $CR = (90-45)/(90-0) = 0.500$, so $\Phi^{-1}(0.500) = 0$, and $Q^* = 500$. Compare with the original Q^* of 530. The absence of salvage value makes overstocking more costly and pulls Q^* to the mean.

Exercise 2: High-Margin vs. Thin-Margin Products

Compare two products, both with $\mu = 500$, $\sigma = 120$:

	p	c	s	CR	Risk Profile
Product A	\$200	\$100	\$20	0.556	High margin, high risk
Product B	\$60	\$50	\$40	0.500	Thin margin, low risk

Run the simulation for each. Product A has higher absolute profit but also higher standard deviation and more negative VaR at extreme Q values. Product B has minimal upside but almost no downside. This demonstrates how margin structure shapes the risk profile.

Exercise 3: AI/ML Extension

In practice, μ and σ are estimated from data. A gradient boosted model trained on 3 years of weekly sales data with features (seasonality, promotions, weather, competitor pricing) might produce $\hat{\mu} = 520$ with a 90% prediction interval of [380, 660], implying $\hat{\sigma} \approx 85$ (compared to the historical $\sigma = 120$). Feed these ML-derived parameters into the dashboard and observe that Q^* shifts and VaR improves.

Discussion: Why does a better demand forecast reduce both σ and the gap between $\mathbb{E}[\pi]$ at Q^* versus $Q = \mu$?

4 Curriculum Alignment

4.1 Mapping to BU Administrative Sciences

The three modules directly support multiple program areas within the Administrative Sciences department at Boston University Metropolitan College. Module 1 bridges supply chain management and financial management. Module 2 introduces financial instruments at the intersection of SCM and enterprise risk management. Module 3 integrates applied business analytics with operations management via Monte Carlo simulation and financial risk metrics.

Module	Primary Course Area	Secondary	Session
CCC & Working Capital	Financial Management	Supply Chain Mgmt	75–90 min
Dynamic Discounting & RF	Supply Chain Management	Risk Management	75–90 min
Monte Carlo Profit Risk	Business Analytics / AI-ML	Operations Mgmt	90–120 min
Integrated Dashboard	Capstone / Integrative	All of the above	2–3 hours

4.2 AI and Machine Learning Extensions

Each module includes a natural extension point for AI/ML. In Module 1, predictive models can forecast DIO, DSO, and DPO from operational data for proactive working capital management. In Module 2, dynamic pricing algorithms optimize the discount APR in real time based on buyer cash position and supplier urgency. In Module 3, the Monte Carlo framework directly consumes probabilistic demand forecasts from ML models, replacing the assumed normal distribution with empirical distributions that capture nonlinearities, seasonality, and external covariates.

4.3 Technical Notes

The dashboard is a single HTML file (∼40 KB) with embedded CSS and JavaScript. Visualizations use the HTML5 Canvas 2D rendering API. Monte Carlo simulation uses the Box-Muller transform for normal random variates and runs 10,000 trials in under 100 milliseconds. The inverse normal CDF uses the Abramowitz and Stegun rational approximation. All computation is client-side, ensuring data privacy and offline capability.

4.4 Suggested Readings

- Brealey, R. A., Myers, S. C., & Allen, F. (2020). *Principles of Corporate Finance* (13th ed.). McGraw-Hill. Chapters 29–30.

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- Simchi-Levi, D., Kaminsky, P., & Simchi-Levi, E. (2021). *Designing and Managing the Supply Chain* (4th ed.). McGraw-Hill.